

Ava is hiking a section of the Citrus Hiking Trail in Withlacoochee State Forest. Ava packs trail mix and grapes for snacks. The trail mix provides 5 calories per ounce and the grapes provide 60 calories. The function  $C(x) = 5x + 60$  where  $x$  is the number of ounces of trail mix Ava eats and  $C$  is the total number of calories she consumes. Ava does not want to consume more than 115 calories.

1. Solve  $C(x) = 115$ .

$$115 = 5x + 60$$

$$-60 \quad -60$$

$$\frac{55}{5} = \frac{5x}{5}$$

$$x = 11$$

2. Write the range of the function  $C(x) = 5x + 60$  using inequalities.

$$0 \leq y \leq 115$$

3. Determine the  $x$ -intercept.

$$0 = 5x + 60$$

$$\frac{-60}{5} = \frac{5x}{5}$$

$$x = -12$$

$$(-12, 0)$$

4. Interpret the  $x$ -intercept.

Since  $x$  represents the number of ounces of trail mix, it must be a positive number. In this scenario, the  $x$ -intercept would tell us she would need to eat  $-12$  ounces of trail mix to consume 0 calories. It is not a viable answer in the context of this scenario.

5. Write the domain of the function  $C(x) = 5x + 60$  using inequalities.

$$0 \leq x \leq 11$$

6. Determine the  $y$ -intercept.

$$(0, 60)$$

7. Interpret the  $y$ -intercept.

If Ava eats 0 ounces of trail mix, she will consume 60 calories from the grapes.

8. Determine the slope.

$$5$$

9. Interpret the slope.  
Ava consumes 5 calories per ounce of trail mix.

10. Graph the function.

